Trạng thái	Đã xong
Bắt đầu vào lúc	Thứ Năm, 7 tháng 11 2024, 4:17 PM
Kết thúc lúc	Thứ Năm, 7 tháng 11 2024, 8:05 PM
Thời gian thực hiện	3 giờ 48 phút
Điểm	11,00/11,00
Điểm	**10,00** trên 10,00 (**100**%)

Câu hỏi **1**

Đúng

Đạt điểm 1,00 trên 1,00

Write and run a program that gives the user only three choices: convert from Fahrenheit to Celsius, convert from Celsius to Fahrenheit, or quit. If the third choice is selected, the program stops. If one of the first two choices is selected, the program should prompt the user for either a Fahrenheit or Celsius temperature, as appropriate, and then compute and display a corresponding temperature. Use the conversion formulas:

- F = (9.0 / 5) . C + 32
- C = (5.0 / 9) . (F - 32)

For example:

Input	Result
1 72.4	22.4444
2 35.3	95.54
12 49	Invalid option

Answer: (penalty regime: 0, 0, 0, 10, 20, ... %)

Reset answer

```cpp
#include<iostream>
using namespace std;
#include<iomanip>
int main() {
    //TODO


    int choice;
    double temperature,f,c;
    cin >> choice;  // choice=1,2
    cin >> temperature;
    if ( choice == 1) {
        c = (5.0 / 9)*(temperature - 32);
        cout << fixed<< setprecision(4)<< c;
    }
    if (choice ==2 ){
        f= (9.0/5)*temperature+32;
        cout << fixed<< setprecision(2)<< f;
    }
    if (choice !=1 & choice !=2) cout << "Invalid option"<<endl;


    return 0;

}
```

	Input	Expected	Got	
✓	1 72.4	22.4444	22.4444	✓
✓	2 35.3	95.54	95.54	✓
✓	12 49	Invalid option	Invalid option	✓
✓	1 78	25.5556	25.5556	✓
✓	2 28.4	83.12	83.12	✓

Passed all tests! ✓

▾ Show/hide question author's solution (Cpp)

```cpp
int main() {
    int option;
```

```
    double temperature;
    cin >> option;
    cin >> temperature;
    if(option == 1) {
        cout << (5/9.0)*(temperature - 32);
    } else if (option == 2) {
        cout << (9/5.0)*temperature + 32;
    } else {
        cout << "Invalid option";
    }
    return 0;
}
```

Đúng

Marks for this submission: 1,00/1,00.

Câu hỏi **2**

Đúng

Đạt điểm 1,00 trên 1,00

Write a program to calculate the number of results obtained when solving the quadratic equation: $ax^2 + bx + c = 0$ with given real inputs *a*, *b*, and *c*.

For example:

Input	Result
`0 0 0`	`Countless solutions`
`0 0 -4`	`No solution`
`0 2 1`	`-0.5000`
`1 2 1`	`-1.0000`
`1 2 2`	`-1.0000 + 1.0000*i` `-1.0000 - 1.0000*i`

Answer: (penalty regime: 0 %)

Reset answer

```
include <iostream>
include <cmath>
include <complex>
include <iomanip>

sing namespace std;

nt main() {
   double a, b, c;
   cin >> a >> b >> c;

   double delta = b*b - 4*a*c;

   if (a == 0) {
       if (b == 0) {
           if (c == 0) {
               cout << "Countless solutions" << endl;
           } else {
               cout << "No solution" << endl;
           }
       } else {
           cout  << fixed << setprecision(4) << -c/b << endl;
       }
   } else {
       if (delta > 0) {
           double x1 = (-b + sqrt(delta)) / (2*a);
           double x2 = (-b - sqrt(delta)) / (2*a);
           cout <<  fixed << setprecision(4) << x1 << endl;
           cout<< fixed << setprecision(4) << x2 << endl;
       } else if (delta == 0) {
           double x = -b / (2*a);
           cout << fixed << setprecision(4) << x << endl;
       } else {
           complex<double> x1 = (-b + complex<double>(0, sqrt(-delta))) / (2*a);
           complex<double> x2 = (-b - complex<double>(0, sqrt(-delta))) / (2*a);
          cout << fixed << setprecision(4) << x1.real() << " + " << fixed << setprecision(4) << x1.imag()<<"*
           cout << fixed << setprecision(4) << x2.real() << " - " << fixed << setprecision(4) << -x2.imag()<<
       }
   }

   return 0;

```

	Input	Expected	Got	
✓	0 0 0	Countless solutions	Countless solutions	✓
✓	0 0 -4	No solution	No solution	✓
✓	0 2 1	-0.5000	-0.5000	✓
✓	1 2 1	-1.0000	-1.0000	✓
✓	1 2 2	-1.0000 + 1.0000*i -1.0000 - 1.0000*i	-1.0000 + 1.0000*i -1.0000 - 1.0000*i	✓
✓	13.12 -9.9 281892126.110	0.3773 + 4635.2648*i 0.3773 - 4635.2648*i	0.3773 + 4635.2648*i 0.3773 - 4635.2648*i	✓
✓	135.12 19.4 122.1	-0.0718 + 0.9479*i -0.0718 - 0.9479*i	-0.0718 + 0.9479*i -0.0718 - 0.9479*i	✓

Passed all tests! ✓

▾ Show/hide question author's solution (Cpp)

```cpp
int main() {
    double a,b,c;
    cin >> a >> b >> c;
    cout << setprecision(4) << fixed;
    if (a == 0) {
        if (b == 0) {
            if (c == 0) {
                cout << "Countless solutions";
            }
            else cout << "No solution";
        } else cout << -c / (double)b;
    } else {
        double delta = b*b - 4*a*c;
        if(delta == 0) {
            cout << -b/(2*a);
        } else if (delta > 0) {
            cout << -b + sqrt(delta) / (2*a) << endl;
            cout << -b - sqrt(delta) / (2*a);
        } else {
            cout << -b / (2*a) << " + " << sqrt(-delta) / (2*a) << "*i" << endl;
            cout << -b / (2*a) << " - " << sqrt(-delta) / (2*a) << "*i";
        }
    }
    return 0;
}
```

Đúng

Marks for this submission: 1,00/1,00.

Câu hỏi **3**

Đúng

Đạt điểm 1,00 trên 1,00

Write and run a program that reads an angle (expressed in degrees) and states in which quadrant the given angle lies. An angle A is said to be in the

- first quadrant if it is in the range 0 <= A < 90
- second quadrant if it is in the range 90 <= A < 180
- third quadrant if it is in the range 180 <= A < 270
- fourth quadrant if it is in the range 270 <= A < 360.

Example
Input: 47
Output: first quadrant

Input: 235.5
Output: third quadrant

Input: 380
Output: not exist

For example:

Test	Input	Result
47	47	first quadrant
235.5	235.5	third quadrant
380	380	not exist

Answer: (penalty regime: 0 %)

Reset answer

```
#include <iostream>
#include <cmath>
using namespace std;
int main()
{
    float angel;
    cin >> angel;
    if( angel<90 and  angel >=0) cout<< "first quadrant";
    if( angel<180 and angel >=90) cout<< "second quadrant";
    if( angel<270 and angel >=180) cout<< "third quadrant";
    if( angel<360 and angel >=270) cout<< "fourth quadrant";
    if ( angel >= 360) cout <<"not exist";
    if ( angel < 0) cout <<"not exist";

    return 0;
}
```

	Test	Input	Expected	Got	
✓	47	47	first quadrant	first quadrant	✓
✓	235.5	235.5	third quadrant	third quadrant	✓
✓	380	380	not exist	not exist	✓
✓	-15	-15	not exist	not exist	✓

	Test	Input	Expected	Got	
✓	360	360	not exist	not exist	✓
✓	90.5	90.5	second quadrant	second quadrant	✓
✓	180.7	180.7	third quadrant	third quadrant	✓
✓	360.1	360.1	not exist	not exist	✓
✓	-0.5	-0.5	not exist	not exist	✓
✓	270.9	270.9	fourth quadrant	fourth quadrant	✓

Passed all tests! ✓

▾ Show/hide question author's solution (Cpp)

```cpp
int main()
{
    double angle;
    cin >> angle;
    if (angle  >= 0 && angle < 90 ) {
        cout << "first quadrant";
    }
    else if (angle >= 90 && angle < 180) {
        cout << "second quadrant";
    }
    else if (angle >= 180 && angle < 270) {
        cout << "third quadrant";
    }
    else if (angle >= 270 && angle < 360)
    {
        cout << "fourth quadrant";
    }
    else
    {
         cout << "not exist";
    }
    return 0;
}
```

Đúng

Marks for this submission: 1,00/1,00.

Câu hỏi **4**

Đúng

Đạt điểm 1,00 trên 1,00

Write and run a program that reads a marriage code (one character) and writes out a message corresponding to the character:

- if the character is ‘M’ or ‘m’ then the message is “Individual is married”
- if the character is ‘D’ or ‘d’ then the message is “Individual is divorced”
- if the character is ‘W’ or ‘w’ then the message is “Individual is widowed”
- otherwise, the message is “An invalid code was entered”.

(Hint: use *switch* statement).

.

For example:

Input	Result
`M`	`Individual is married`
`c`	`An invalid code was entered`

Answer: (penalty regime: 0. %)

Reset answer

```
# include <iostream>
using namespace std;


int main(){
    //TODO
    char x;
    bool check=true;
    cin >>x;
    if (x =='M'|| x =='m') {
        cout<< "Individual is married";
        check = false;
    }
    if (x =='D'|| x =='d') {
        cout<< "Individual is divorced";
        check= false;
    }
    if (x =='W'||x =='w') {
        cout<< "Individual is widowed";
        check = false;
    }
    if (check) cout<<"An invalid code was entered";





    return  0;
}
```

	Input	Expected	Got	
✓	`M`	`Individual is married`	`Individual is married`	✓
✓	`c`	`An invalid code was entered`	`An invalid code was entered`	✓
✓	`m`	`Individual is married`	`Individual is married`	✓
✓	`D`	`Individual is divorced`	`Individual is divorced`	✓
✓	`d`	`Individual is divorced`	`Individual is divorced`	✓

	Input	Expected	Got	
✓	w	Individual is widowed	Individual is widowed	✓
✓	W	Individual is widowed	Individual is widowed	✓
✓	o	An invalid code was entered	An invalid code was entered	✓

Passed all tests! ✓

▾ Show/hide question author's solution (Cpp)

```cpp
int main(){
    char marriage_code;
    cin >> marriage_code;
    switch(marriage_code){
        case 'M':
        case 'm':
            cout << "Individual is married";
            break;
        case 'D':
        case 'd':
            cout << "Individual is divorced";
            break;
        case 'W':
        case 'w':
            cout << "Individual is widowed";
            break;
        default:
            cout << "An invalid code was entered";
    }
    return 0;
}
```

Đúng

Marks for this submission: 1,00/1,00.

Câu hỏi **5**

Đúng

Đạt điểm 1,00 trên 1,00

Write C++ program to find the largest number among three numbers.

Example

Input: 3.5 -6.7 8.5

Output: 8.5

For example:

Input	Result
3.65 4.78 5.23	5.23

Answer: (penalty regime: 0 %)

Reset answer

```
#include<iostream>
using namespace std;
int main(){
    //TODO
    float x,y,z,max;

    cin>>x>>y>>z;
    max= x;
    if ( max<y) max=y;
    if(max<z) max =z;
    cout <<max;
    return 0;
}
```

	Input	Expected	Got	
✓	3.65 4.78 5.23	5.23	5.23	✓
✓	-10 -17 -14	-10	-10	✓
✓	23.5 10.6 11	23.5	23.5	✓
✓	4.6 4.7 4.8	4.8	4.8	✓

Passed all tests! ✓

▾ Show/hide question author's solution (Cpp)

```
int main()
{
    float n1, n2, n3;
    cin >> n1 >> n2 >> n3;

    if(n1 >= n2 && n1 >= n3)
        cout << n1;

    if(n2 >= n1 && n2 >= n3)
        cout << n2;

    if(n3 >= n1 && n3 >= n2)
```

```
    if(n3 >= n1 && n3 >= n2)
        cout << n3;

    return 0;
}
```

Đúng

Marks for this submission: 1,00/1,00.

Câu hỏi **6**

Đúng

Đạt điểm 1,00 trên 1,00

Write and run a program that reads a month from the user and displays the number of days in that month.

Example:

Input: 3
Output: 31

Input: 2
Output: 28 or 29

For example:

Input	Result
3	31
2	28 or 29

Answer: (penalty regime: 0, 0, 0, 10, 20, ... %)

Reset answer

```
#include<iostream>
using namespace std;
int main()
{
    //TODO
    int month;
    cin>>month;
    switch ( month){
        case 1:
        cout <<"31";
        break;
        case 2:
        cout<<"28 or 29";
        break;
        case 3:
        cout <<"31";
        break;
        case 4:
        cout<<"30";
        break;
        case 5:
        cout<<"31";
        break;
        case 6:
        cout<<"30";
        break;
        case 7:
        cout<<"31";
        break;
        case 8:
        cout<<"31";
        break;
        case 9:
        cout<<"30";
        break;
        case 10:
        cout <<"31";
        break;
        case 11:
        cout<<"30";
        break;
        case 12:
        cout<<"31";
        break;
    }
    return 0;
    }
```

	Input	Expected	Got	
✓	3	31	31	✓
✓	2	28 or 29	28 or 29	✓
✓	1	31	31	✓
✓	7	31	31	✓
✓	9	30	30	✓

Passed all tests! ✓

▾ Show/hide question author's solution (Cpp)

```
int main()
{
    int month;
    cin >> month;
    if (month % 2 == 0){
        if(month == 2) {
            cout << "28 or 29";
        } else if (month <= 6) {
            cout << 30;
        } else {
            cout << 31;
        }
    } else {
        if(month <= 7) {
            cout << 31;
        } else cout << 30;
    }
    return 0;
}
```

Đúng

Marks for this submission: 1,00/1,00.

Câu hỏi **7**

Đúng

Đạt điểm 1,00 trên 1,00

Write C++ program to checks whether an year (integer) entered by the user is a leap year or not.

For example:

Input	Result
2000	2000 is a leap year.
2002	2002 is not a leap year.

Answer: (penalty regime: 0 %)

Reset answer

```
#include<iostream>
using namespace std;
int main(){
    //TODO
    int year;
    cin>>year;
    if ((year % 4 == 0 )&( year % 100 !=0) ||( year % 400 == 0)) cout<<year<<" is a leap year.";
    else cout<<year<<" is not a leap year.";
    return 0;
}
```

	Input	Expected	Got	
✓	2000	2000 is a leap year.	2000 is a leap year.	✓
✓	2002	2002 is not a leap year.	2002 is not a leap year.	✓
✓	1900	1900 is not a leap year.	1900 is not a leap year.	✓
✓	1935	1935 is not a leap year.	1935 is not a leap year.	✓
✓	1700	1700 is not a leap year.	1700 is not a leap year.	✓

Passed all tests! ✓

▾ Show/hide question author's solution (Cpp)

```
int main(){
    int year;
    cin >> year;
    if (year % 4 == 0) {
        if (year % 100 == 0) {
            if (year % 400 == 0)
                cout << year << " is a leap year.";
            else
                cout << year << " is not a leap year.";
        }
        else
            cout << year << " is a leap year.";
```

```
            cout << year << " is a leap year.";
    }
    else
        cout << year << " is not a leap year.";
    return 0;
}
```

Đúng

Marks for this submission: 1,00/1,00.

Câu hỏi **8**

Đúng

Đạt điểm 1,00 trên 1,00

Write C++ program to check whether a number entered by the user is even or odd.

For example:

Input	Result
20	20 is even.
-5	-5 is odd.

Answer: (penalty regime: 0 %)

Reset answer

```
#include<iostream>
using namespace std;
int main(){
    //TODO
    int x;
    cin>>x;
    if(x %2 == 0) cout <<x<<" is even.";
    else cout<<x<<" is odd.";
    return 0;
}
```

	Input	Expected	Got	
✓	20	20 is even.	20 is even.	✓
✓	-5	-5 is odd.	-5 is odd.	✓
✓	14	14 is even.	14 is even.	✓
✓	13	13 is odd.	13 is odd.	✓
✓	-105	-105 is odd.	-105 is odd.	✓

Passed all tests! ✓

▾ Show/hide question author's solution (Cpp)

```
int main()
{
    int n;
    cin >> n;

    if ( n % 2 == 0)
        cout << n << " is even.";
    else
        cout << n << " is odd.";

    return 0;
}

```

Đúng

Marks for this submission: 1,00/1,00.

Câu hỏi 9

Đúng

Đạt điểm 1,00 trên 1,00

Write C++ program to check whether a given number is positive or negative.

For example:

Input	Result
15	15 is a positive number.
-4	-4 is a negative number.

Answer: (penalty regime: 0 %)

Reset answer

```
#include<iostream>
using namespace std;
int main(){
    //TODO
    int x;
    cin>>x;
    if(x>0) cout <<x<<" is a positive number.";
    if(x<0) cout <<x<<" is a negative number.";
    return 0;
}
```

	Input	Expected	Got	
✓	15	15 is a positive number.	15 is a positive number.	✓
✓	-4	-4 is a negative number.	-4 is a negative number.	✓
✓	-9	-9 is a negative number.	-9 is a negative number.	✓
✓	35	35 is a positive number.	35 is a positive number.	✓
✓	-23	-23 is a negative number.	-23 is a negative number.	✓

Passed all tests! ✓

▾ Show/hide question author's solution (Cpp)

```
int main ()
{
    int num;
    cin >> num;
    if (num >= 0)
        cout << num << " is a positive number.";
    else cout << num << " is a negative number.";
    return 0;
}

```

Đúng

Marks for this submission: 1,00/1,00.

Câu hỏi **10**

Đúng

Đạt điểm 1,00 trên 1,00

Write C++ program to check whether a Triangle is Right-angled, Equilateral, Isosceles or Scalene

For example:

Input	Result
4 5 3	The triangle is Right-angled.
3 5 3	The triangle is Isosceles.

Answer: (penalty regime: 0 %)

Reset answer

```
#include <iostream>
#include <cmath>

using namespace std;

int main() {
    int a, b, c;
    cin >> a >> b >> c;

    if (a == b && b == c) {
        cout << "The triangle is Equilateral." << endl;
    } else if (a == b || b == c || c == a) {
        cout << "The triangle is Isosceles." << endl;
    } else if (pow(a, 2) == pow(b, 2) + pow(c, 2) ||
               pow(b, 2) == pow(a, 2) + pow(c, 2) ||
               pow(c, 2) == pow(a, 2) + pow(b, 2)) {
        cout << "The triangle is Right-angled." << endl;
    } else {
        cout << "The triangle is Scalene." << endl;
    }

    return 0;
}
```

	Input	Expected	Got	
✓	4 5 3	The triangle is Right-angled.	The triangle is Right-angled.	✓
✓	3 5 3	The triangle is Isosceles.	The triangle is Isosceles.	✓
✓	3 6 5	The triangle is Scalene.	The triangle is Scalene.	✓
✓	4 4 4	The triangle is Equilateral.	The triangle is Equilateral.	✓
✓	10 6 8	The triangle is Right-angled.	The triangle is Right-angled.	✓

Passed all tests! ✓

Show/hide question author's solution (Cpp)

```
int main(){
    int a,b,c;
    cin >> a >> b >> c;
    if(a==b && b==c && c==a){
        cout << "The triangle is Equilateral.";
    } else if (a==b || b==c || c==a){
        cout << "The triangle is Isosceles.";
    } else if(a*a==b*b+c*c ||b*b==c*c+a*a || c*c==a*a+b*b) {
```

```
        cout << "The triangle is Right-angled.";
    } else {
        cout << "The triangle is Scalene.";
    }
}
```

Đúng

Marks for this submission: 1,00/1,00.

Câu hỏi **11**

Đúng

Đạt điểm 1,00 trên 1,00

Write and run a program that reads a numeric grade from a student and displays a corresponding character grade for that numeric grade. The program prints ‘A’ for exam grades greater than or equal to 90, ‘B’ for grades in the range 80 to 89, ‘C’ for grades in the range 70 to 79, ‘D’ for grades in the range 60 to 69 and ‘F’ for all other grades.

For example:

Input	Result
69.5	D
112	Grade is not exist

Answer: (penalty regime: 0 %)

Reset answer

```
#include<iostream>
using namespace std;
char getGrade(float grade) {
    if (grade >= 90 && grade <=100) return 'A';
    if (grade >= 80 && grade < 90 ) return 'B';
    if (grade >= 70 && grade < 80) return 'C';
    if (grade >= 60 && grade < 70) return 'D';
      if (grade >=0 && grade <60) return 'F';
      return '1';
}
int main(){
    //TODO
    float grade;
    bool check = false;
    cin>>grade;
    char result;
          result=getGrade(grade);
     if (result=='1') {
         cout<<"Grade is not exist";
     }
     else cout<<result;



    return 0;
  }
```

	Input	Expected	Got	
✓	69.5	D	D	✓
✓	112	Grade is not exist	Grade is not exist	✓
✓	-4	Grade is not exist	Grade is not exist	✓
✓	89.5	B	B	✓
✓	92	A	A	✓
✓	49	F	F	✓
✓	76	C	C	✓

Passed all tests! ✓

▾ Show/hide question author's solution (Cpp)

```
int main() {
```

```
    double grade;
    cin >> grade;
    if (grade > 100 || grade < 0){
        printf("Grade is not exist");
    } else if (grade >= 90) {
        printf("A");
    } else if (grade >= 80) {
        printf("B");
    } else if (grade >= 70) {
        printf("C");
    } else if (grade >= 60) {
        printf("D");
    } else printf("F");
    return 0;
}
```

Đúng

Marks for this submission: 1,00/1,00.